Disparity in Transport of Critically Injured Patients to Trauma Centers: Analysis of the National Emergency Medical Services Information System (NEMSIS)

Natalie Escobar, BS, Charles DiMaggio, PhD, Spiros G Frangos, MD, MPH, FACS, Robert J Winchell, MD, FACS, Marko Bukur, MD, FACS, Michael J Klein, MD, FACS, Leandra Krowsoski, MD, FACS, Manish Tandon, MD, FACS, Cherisse Berry, MD, FACS

BACKGROUND: Patient morbidity and mortality decrease when injured patients meeting CDC Field Triage Criteria (FTC) are transported by emergency medical services (EMS) directly to designated trauma centers (TCs). This study aimed to identify potential disparities in the transport of critically injured patients to TCs by EMS.

STUDY DESIGN: We identified all patients in the National EMS Information System (NEMSIS) database in the National Association of EMS State Officials East region from January 1, 2018, to December 31, 2019, with a final prehospital acuity of critical or emergent by EMS. The cohort was stratified into patients transported to TCs or non-TCs. Analyses consisted of descriptive epidemiology, comparisons, and multivariable logistic regression analysis to measure the association of demographic features, vital signs, and CDC FTC designation by EMS with transport to a TC.

RESULTS: A total of 670,264 patients were identified as sustaining an injury, of which 94,250 (14%) were critically injured. Of those 94,250 critically injured, 56.0% (52,747) were transported to TCs. Among all critically injured women (n = 41,522), 50.4% were transported to TCs compared with 60.4% of critically injured men (n = 52,728, $p < 0.001$). In a multivariable logistic regression model, critically injured women were 19% less likely to be taken to a TC compared with critically injured men (OR 0.81, 95% CI 0.71–0.93, $p = 0.003$).

CONCLUSIONS: Critically injured female patients are less likely to be transported to TCs when compared with their male counterparts. Performance improvement processes that assess EMS compliance with field triage guidelines should explicitly evaluate for sex-based disparities. Further studies are warranted. (J Am Coll Surg 2022;235:78–85. © 2022 by the American College of Surgeons. Published by Wolters Kluwer Health, Inc. All rights reserved.)

Unintentional injury is the leading cause of death for Americans aged 1 to 44 years.[1] When patients are critically injured, rapid assessment, scene-time stabilization, and transport to hospitals for definitive care are essential in achieving zero preventable deaths. CDC Field Triage Criteria (FTC) were developed by the American College of Surgeons' Committee on Trauma (ACS COT) and the National Highway Traffic Safety Administration to provide a standardized approach for emergency medical services (EMS) to recognize seriously injured patients who would benefit from transport to a designated/verified trauma center (TC). This standardized approach involves an assessment of not only the physiology and anatomy of the injury but also the mechanism of the injury and consideration of special populations.[2] Patient morbidity and mortality decrease when injured patients meeting CDC

Disclosure Information: Dr Bukur is a paid consultant for Zimmer Biomet.

Presented at the Western Surgical Association 129th Scientific Session, Indian Wells, CA, November 2021.

Received January 11, 2022; Revised February 11, 2022; Accepted February 15, 2022.

From the NYC Health & Hospitals-Bellevue-Department of Surgery, New York University Grossman School of Medicine, New York, NY (Escobar, DiMaggio, Frangos, Bukur, Klein, Krowsoski, Tandon, Berry); Department of Surgery, Weill Cornell Medical College, New York, NY (Winchell).

Correspondence address: Cherisse Berry, MD, FACS, Department of Surgery, Division of Acute Care Surgery, New York University Grossman School of Medicine, 550 First Ave, NBV 12 East 36, New York, NY 10016. email: cherisse.berry@nyulangone.org

A discussion of this article is available at http://links.lww.com/JACS/A69.

https://doi.org/10.1097/XCS.0000000000000230
ISSN 1879-1190/22

Abbreviations and Acronyms

ACS COT	= American College of Surgeons' Committee on Trauma
EMS	= emergency medical services
FTC	= Field Triage Criteria
NEMSIS	= National EMS Information System
TC	= trauma center

FTC are transported by EMS directly to designated TCs.[3,4] Although previous studies have demonstrated a marked survival benefit when injured patients are transported to a TC vs non-TC,[4–9] no study to date has evaluated factors such as sex disparities that influence the decision by EMS to transport injured patients to TCs. Based on previous data,[10,11] we sought to identify sex-based disparities in the transport of injured patients to TCs. We hypothesize that sex-based differences in the transport of critically injured patients is prevalent.

METHODS

This study was a retrospective review of the National EMS Information System (NEMSIS) database from January 1, 2018, to December 31, 2019. Data from the National Association of EMS State Officials (NASEMO) East region including Connecticut, Delaware, Maine, Massachusetts, Maryland and the District of Columbia, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and West Virginia with a final transport destination recorded were identified. Patients with an injury and a final acuity designated by EMS as critical or emergent were included. EMS defines critical acuity as a life-threatening illness or injury with a high probability of mortality if immediate intervention is not initiated to prevent further airway, respiratory, hemodynamic, and/or neurologic instability, and defines emergent acuity as an illness or injury that may progress in severity or result in complications with a high probability for morbidity if treatment is not administered promptly.[12] Patients dead on the scene, with a final acuity designated by EMS as low, or patients with missing data were excluded. The cohort of critically or emergently injured patients was further stratified into 2 subgroups: (1) critically injured patients designated as meeting CDC FTC by EMS (CDC FTC subgroup), and (2) a field vital sign abnormality subgroup of critically injured patients with a field vital sign abnormality, including a Glasgow Coma Scale at or below 13, systolic blood pressure less than 90 mmHg, or respiratory rate less than 10 or more than 29 breaths per minute (Figure 1). Although the field vital sign abnormality should guarantee both designation as meeting CDC FTC and transport to a trauma center, not all patients with field vital signs abnormalities were correctly designated by EMS as meeting CDC FTC; thus, this cohort identifies patients with vital sign abnormalities regardless of their CDC FTC designation by EMS. TCs were identified using final EMS transport destination zip codes that matched reported zip codes of verified TCs on the websites of the ACS COT, Maryland Institute for Emergency Medical Services Systems, and Pennsylvania Trauma Systems Foundation. Final EMS transport destination zip codes that did not match verified TCs on the above websites were identified as non-TCs. Patient demographics including age, race, sex, insurance status, injury type (blunt vs penetrating), vital signs, EMS response time, scene time, transport time, and EMS reason for final transport destination choice were described for the critically injured cohort and the 2 subgroups (CDC FTC subgroup and field vital sign abnormality subgroup), then compared between the two final transport destination cohorts (TC vs non-TC).

Statistical analysis

Differences between continuously distributed data were tested using *t*-tests; categorical variables were compared using chi-squared tests. Simple logistic regression analysis was performed to determine the univariate association of sex with the designation of meeting CDC FTC by EMS. Multivariable logistic regression analysis with 95% CIs was performed to determine the association of demographics, vital signs, CDC FTC designation by EMS, and injury on the transport to designated/verified TCs. Confounding variables such as age, sex, race, insurance status, CDC FTC, penetrating/blunt injury, urbanicity, and vital signs were included based on a priori consideration of factors believed to be associated with the transport to a TC. R statistical software was used to analyze the data, and a p value less than 0.05 was set for statistical significance. The study was determined to be exempt by the Institutional Review Board of New York University Grossman School of Medicine.

RESULTS

A total of 94,250 injured patients with a final acuity of critical or emergent met the inclusion criteria. The mean age was 52.7 years, and the majority of the patients were male (55.9%, n = 52,728), unknown race/other (61.5%, n = 57,989), and insured (46.3%, n =10,827); 56% (n = 52,747) of the total cohort were transported to a TC (Table 1). The most common reason for destination chosen by EMS was closest facility (41.6%, n = 23,666;

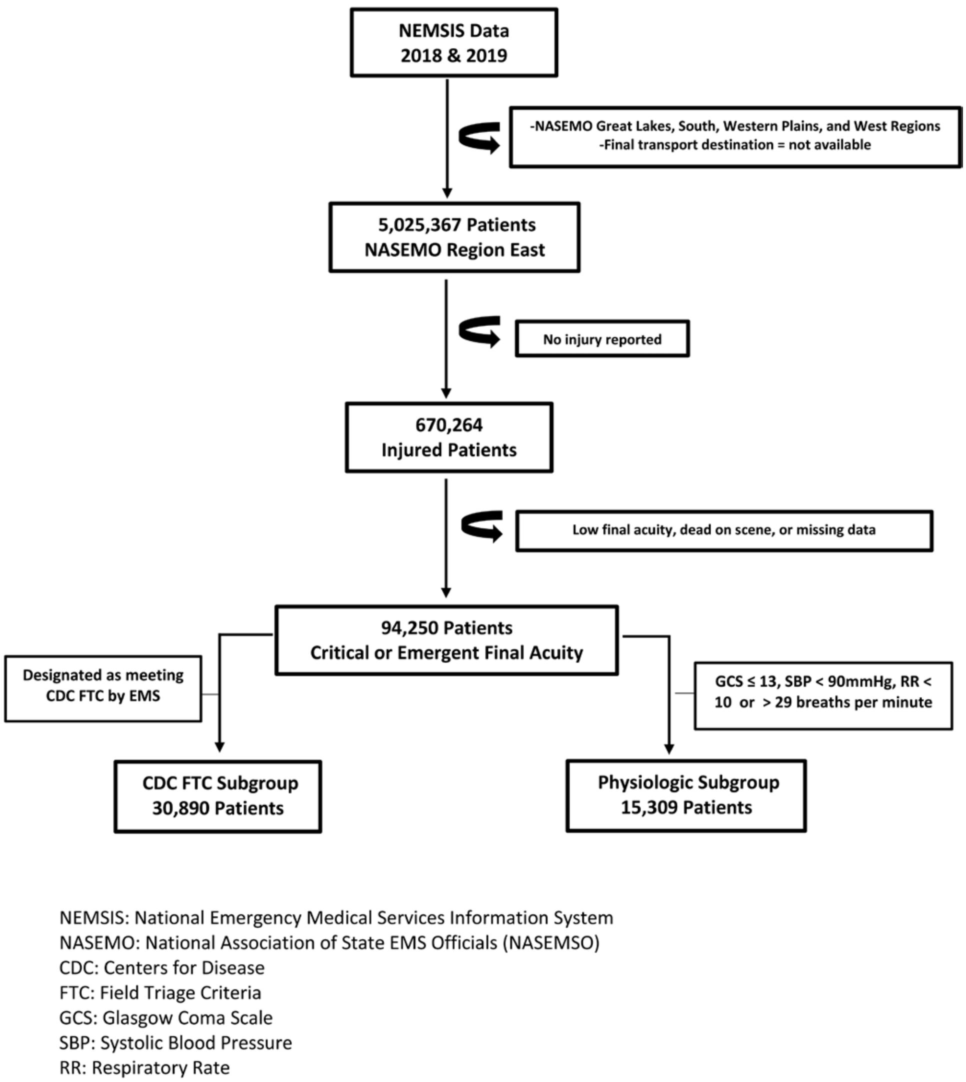

Figure 1. Flow chart of patient cohort.

Table 2). In a multivariable logistic regression analysis that included all critically injured patients (n = 94,250), women were 19% less likely to be taken to a TC compared with critically injured men (odds ratio [OR] 0.81, 95% CI 0.71–0.93, p = 0.003, AUC 0.71; Table 3).

CDC FTC subgroup

Of the 94,250 critically injured patients, 30,890 (57.5%) were designated as meeting CDC FTC by EMS (CDC FTC subgroup). Although 76.5% (n = 23,642) of this subgroup were transported to a TC, when stratified by sex, a larger percentage of men were transported to TCs when compared with their female counterparts (77.9% vs 74.1%, p < 0.001; Table 2). However, on multivariable logistic regression analysis, although designation as meeting CDC FTC increased the odds of critically injured patients being transported to a TC (OR 1.69, 95% CI 1.47–1.93, p < 0.001, AUC 0.71), there was no relationship between sex and the transport to a TC within this subgroup (Table 3).

Field vital sign abnormality subgroup

Of the 15,309 critically injured with field vital sign abnormalities, only 44.6% (n = 6,834) were designated by EMS as meeting CDC FTC. Of those, 74.9% (n = 5,117) were transported to designated/verified TCs. When stratified by sex, a higher percentage of males were transported to TCs when compared with their female counterparts (76% vs 72%, p = 0.001). Of the 15,309 patients with unstable vitals, 18.3% (n = 2,802) were not designated

Table 1. Demographic and Physiologic Variables Stratified by Sex

Patient characteristic	Total n = 94,250	Female n = 41,522	Male n = 52,728	p Value
Age, y, mean ± SD	52.7 ± 24.9	58.1 ± 25.7	48.5 ± 23.4	<0.001
Race, n (%)				<0.001
American Indian/Alaskan Native	268 (0.3)	114 (0.3)	154 (0.3)	
Asian	474 (0.5)	210 (0.5)	264 (0.5)	
Black	7,939 (8.4)	3,307 (8.0)	4,632 (8.8)	
Hispanic/Latino	2,857 (3.0)	1,177 (2.8)	1,680 (3.2)	
Native Hawaiian/other Pacific Islander	88 (0.1)	38 (0.1)	50 (0.1)	
White	57,989 (61.5)	25,726 (62.0)	32,263 (61.2)	
Other	24,635 (26.1)	10,950 (26.4)	13,685 (26.0)	
Insurance, n (%)				<0.001
Insured	6,537 (28.0)	3,069 (30.6)	3,468 (26.1)	
Medicaid	1,687 (7.2)	714 (7.1)	973 (7.3)	
Medicare	4,290 (18.4)	2,434 (24.3)	1,856 (13.9)	
No insurance identified	7,774 (33.3)	2,864 (28.6)	4,910 (36.9)	
Self-pay	3,049 (13.1)	946 (9.4)	2,103 (15.8)	
Total (n)	23,337	10,027	13,310	
Vitals, mean ± SD				
Systolic blood pressure, mmHg	139.74 ± 27.3	141.16 ± 27.9	138.61 ± 26.9	<0.001
Heart rate, beats per minute	88.84 ± 21.1	88.58 ± 19.9	89.04 ± 22.0	0.001
GCS	13.95 ± 2.6	14.20 ± 2.2	13.76 ± 2.9	<0.001
Respiratory rate, breaths per minute	18.26 ± 4.6	18.27 ± 4.4	18.24 ± 4.8	0.36
Transported to trauma center, n (%)				<0.001
No	41,503 (44.0)	20,598 (49.6)	20,905 (39.6)	
Yes	52,747 (56.0)	20,924 (50.4)	31,823 (60.4)	

GCS, Glasgow Coma Scale score.

by EMS as meeting CDC FTC. Of those, 53.9% (n = 1,510) were taken to TCs. When stratified by sex, a higher percentage of males were transported to TCs when compared with their female counterparts (58.1% vs 48%, p < 0.001). Designation by EMS as meeting CDC FTC was not recorded in 37% (n = 5,673) of the cohort; of these, a higher percentage of males were transported to TCs when compared with their female counterparts (49.2% vs 44.3%, p < 0.001; Table 2). On simple logistic regression, women were 38.9% less likely to be designated as meeting CDC FTC by EMS when compared with men with similar vitals (OR 0.72, 95% CI 0.53–0.98, p < 0.001). On multivariable logistic regression analysis, women were 26% less likely to be taken to a TC (OR 0.74, 95% CI 0.55–0.997, p = 0.046, AUC 0.66; Table 3).

DISCUSSION

Transport to verified TCs is associated with a significant reduction in preventable deaths and injury-related mortality.[2,5–9] To our knowledge, this is the largest study to date to demonstrate potential sex-based disparities in the transport of critically injured patients to designated TCs within the US. Despite the presence of standardized, national field triage guidelines, our findings showed critically injured women were less likely to be transported to TCs when compared with critically injured men.

It is well established that racial/ethnic disparities in trauma care exist[13] not only in access[14] to care but also at each point along the trauma care chain of survival[15]: bystander care,[16] prehospital EMS care,[17] hospital definitive care,[18–20] rehabilitation,[21,22] and recovery/re-entry.[23,24] Yet, sex disparities based the current standardized CDC FTC within a prehospital EMS system have not been reported in the US.

The rapid response, assessment, initial stabilization, and transport to designated/verified TCs by EMS has been vital in improving mortality outcomes after trauma.[25,26] The decision to transport seriously injured trauma patients to designated/verified TCs is based on CDC FTC. Previously described deviations in adherence to triage guidelines include patients declining guideline-directed hospital

Table 2. Emergency Medical Services Prehospital Variables Stratified by Sex

Patient characteristic	Total	Female	Male	p Value
CDC field triage criteria				
No, n (%)	22,812 (42.5)	10,877 (49.0)	11,935 (37.9)	<0.001
Yes, n (%)	30,890 (57.5)	11,309 (51.0)	19,581 (62.1)	
Total, n	53,702	22,186	31,516	
Physiologic + anatomic criteria				<0.001
No, n (%)	21,042 (45.7)	9,125 (48.4)	11,917 (43.8)	
Yes, n (%)	25,034 (54.3)	9,732 (51.6)	15,302 (56.2)	
Total, n	46,076	18,857	27,219	
Mechanism + special populations criteria				<0.001
No, n (%)	38,607 (77.2)	17,399 (84.1)	21,208 (72.3)	
Yes, n (%)	11,390 (22.8)	3,283 (15.9)	8,107 (27.7)	
Total, n	49,997	20,682	29,315	
EMS system response time, min, mean ± SD	9.3 ± 16.7	9.3 ± 17.8	9.3 ± 15.8	0.918
EMS scene time, min, mean ± SD	18.4 ± 14.3	18.9 ± 12.4	18.1 ± 15.6	<0.001
EMS transport time, min, mean ± SD	17.0 ± 17.4	16.8 ± 16.9	17.0 ± 17.8	0.096
Destination reason				<0.001
Closest facility, n (%)	23,666 (41.6)	10,614 (42.6)	13,052 (40.9)	
Diversion, n (%)	341 (0.6)	157 (0.6)	184 (0.6)	
Insurance status, n (%)	16 (0.0)	7 (0.0)	9 (0.0)	
Law enforcement, n (%)	45 (0.1)	10 (0.0)	35 (0.1)	
Multiple reasons selected by EMS, n (%)	3,966 (7.0)	1,720 (6.9)	2,246 (7.0)	
Online/on-scene medical direction, n (%)	1,081 (1.9)	440 (1.8)	641 (2.0)	
Other, n (%)	996 (1.8)	389 (1.6)	607 (1.9)	
Patient or family choice, n (%)	11,751 (20.7)	6,096 (24.5)	5,55 (17.7)	
Protocol, n (%)	5,788 (10.2)	2,306 (9.3)	3,482 (10.9)	
Regional specialty center, n (%)	9,216 (16.2)	3,186 (12.8)	6,030 (18.9)	
Total, n	56,866	24,925	31,916	

EMS, emergency medical services.

selection or patient choice, deficits in EMS provider training, hospital proximity, local resource constraints, geographic location, and provider decision-making.[27–29] Our study found the most common reason for destination chosen by EMS was closest facility (44.5%), followed by patient or family choice (23.4%).

Our findings of sex-based disparities in compliance with CDC field triage guidelines is supported by the literature. Rubenson Wahlin and colleagues[10] retrospectively evaluated the trauma registries and ambulance records in Stockholm County, Sweden, and found that female trauma patients were less likely to be given the highest prehospital priority, the highest prehospital competence level, and direct transport to the designated trauma center. Hsia and colleagues[30] retrospectively analyzed the California Office of Statewide Planning and Development patient discharge database to determine the likelihood of admission to a TC for elderly injured patients. In a multivariable logistic regression analysis, female sex was associated with a 22% lower likelihood admission to a TC. Chang and colleagues[31] sought to determine whether age bias was associated with the under-triage of elderly trauma patients to TCs within the statewide Maryland Ambulance Information System database. In a subset analysis focusing on priority 1 patients who met American College of Surgeons triage criteria in Baltimore, they showed that males were significantly more likely (OR 1.36, 95% CI 1.03–1.79, p = 0.03) to be transported to a TC when compared with their female counterparts. Finally, Gomez and colleagues[11] retrospectively evaluated the National Ambulatory Care Reporting System in Ontario, Canada, to evaluate the relationship between sex and access to TCs among severely injured adult patients. Among the 26,861 patients who met the inclusion criteria, sex-based differences in overall access to TCs were prevalent. Women were 13% less likely to receive care in a TC compared with men (OR 0.87, 95% CI 0.79–0.96). Sex-based

Table 3. Multivariable Logistic Regression Analysis to Determine the Association of Demographics, Vital Signs, CDC Field Triage Criteria Designation, and Injury on the Transport to Designated/Verified Trauma Centers

Regression variable	Odds ratio	95% CI	p Value
Critically injured patients (AUC = 0.71)			
Age, y	0.99	0.99–0.99	<0.001
Sex, f	0.81	0.71–0.93	0.003
CDC FTC designation	1.69	1.48–1.94	<0.001
Blunt injury	0.73	0.61–0.88	0.001
Urban	7.26	5.67–9.33	<0.001
Race, White	0.74	0.65–0.86	<0.001
Insured	0.77	0.67–0.89	<0.001
Systolic blood pressure <90 mmHg	0.85	0.59–1.25	0.4
Heart rate >100 beats per min	1.14	0.97–1.34	0.1
GCS ≤3	0.80	0.66–0.99	0.01
Respiratory rate <10 or >29 breaths per min	1.06	0.70–1.67	0.78
Subgroup 2: CDC FTC designation (AUC = 0.65)			
Age, y	0.90	0.99–0.99	0.03
Sex, f	0.96	0.75 –1.20	0.7
Blunt injury	0.60	0.45 –0.79	<0.001
Urban	8.99	6.39 –12.7	<0.001
White	0.67	0.54 –0.83	<0.001
Insured	1.04	0.83 –1.31	0.73
Systolic blood pressure <90 mmHg	1.39	0.79–2.63	0.28
Heart rate >100 beats per min	1.14	0.90–1.45	0.28
GCS ≤13	0.86	0.67 –1.12	0.26
Respiratory rate <10 or >29 breaths per min	1.49	0.76 –3.23	0.27
Subgroup 3: field vital sign abnormality (AUC = 0.66)			
Age, y	0.99	0.99–0.99	0.02
Sex, f	0.74	0.55–0.99	0.05
CDC FTC designation	1.81	1.36–2.42	<0.001
Blunt injury	0.63	0.42–0.94	0.03
Urban	9.57	5.43–17.4	<0.001
Race, White	0.77	0.57–1.03	0.08
Insured	0.70	0.51–0.96	0.02

FTC, field triage criteria; GCS, Glasgow Coma Scale score.

differences in direct transport from the field were prevalent, as women were 12% less likely to be transported to a TC when compared with men (OR 0.88, 95% CI 0.81–0.97). This disparity was also evident in the rate of transfer from non-TCs to TCs, as women were 15% less likely to be transferred when compared with men (OR 0.85, 95% CI 0.73–0.99).

Although the focus of the study was on the presence of sex disparities in the transport of critically injured patients to TCs, our analysis also revealed other concerning findings. Not only were critically injured women less likely to be transported to a TC, but patients with a Glasgow Coma Scale at or below 3 were also 20% less likely to be transported to a TC (OR 0.80, 95% CI 0.66–0.99). Similarly, patients with private insurance or Medicare were 23% less likely to be taken to a verified TC compared with those without insurance or Medicaid (OR 0.77, 95% CI 0.67–0.89).

There are several limitations to our study. First, as with any database study, the accuracy and availability of data relies on voluntarily reported data. Many variables of interest may not be available or accurately reported in the database, such as TC level or patient outcome data. Second, EMS judgment on what constitutes a life-threatening injury may not be accurate or may be overstated in their reporting. Third, whereas previous data have shown a survival benefit to transferring injured patients to verified TCs vs non-TCs, patient outcome data such as morbidity and mortality were not available for this study. Fourth, because of data usage agreements between NEMSIS and individual TCs, we were unable to use geospatial coordinates to accurately identify TCs and non-TCs. We relied on zip code data where there was overlap between the location of about 20% of TCs and non-TCs. Fifth, because race/ethnicity and sex data were not based on patient self-identification but rather on EMS identification, the accuracy of race/ethnicity and the misidentification of patient race or sex as "Unknown/Other" or misgendering of patients, were limitations. Despite these limitations, this is the largest study to date demonstrating an association between sex disparities and the transport of critically injured trauma patients to TCs despite the availability of standardized national field triage guidelines.

CONCLUSIONS

An association between sex disparities and the transport of critically injured trauma patients with TCs exists. Critically injured women are not only less likely to be designated as meeting CDC FTC by EMS when compared with men with similar vital signs, but they are also less likely to be transported to TCs. Performance improvement processes that assess EMS compliance with field triage guidelines should explicitly evaluate for and implement processes aimed to eliminate sex-based disparities. A call to action is needed to specifically analyze EMS prehospital data at the local, regional, state, and national level to ensure that the standards of trauma care are met and health equity is achieved for all patients.

Author Contributions

Study conception and design: Berry, Escobar, DiMaggio
Acquisition of data: Berry, Escobar, DiMaggio
Analysis and interpretation of data: Berry, Escobar, DiMaggio, Frangos, Winchell, Bukur, Tandon, Krowsoski
Drafting of manuscript: Berry, Escobar, DiMaggio, Frangos, Winchell, Bukur, Tandon, Klein, Krowsoski
Critical revision: Berry, Escobar, DiMaggio, Frangos, Winchell, Bukur, Tandon, Klein, Krowsoski

REFERENCES

1. Heron M. Deaths: Leading causes for 2019. NVSS. 2021; Volume 70: Available at: https://www.cdc.gov/nchs/data/nvsr/nvsr70/nvsr70-09-508.pdf. Accessed October 11, 2021.
2. McCoy CE, Chakravarthy B, Lotfipour S. Guidelines for field triage of injured patients: In conjunction with the Morbidity and Mortality Weekly Report published by the Center for Disease Control and Prevention. West J Emerg Med 2013;14:69–76.
3. MacKenzie EJ, Rivara FP, Jurkovich GJ, et al. A national evaluation of the effect of trauma-center care on mortality. N Engl J Med 2006;354:366–378.
4. Haas B, Gomez D, Zagorski B, et al. Survival of the fittest: The hidden cost of undertriage of major trauma. J Am Coll Surg 2010;211:804–811.
5. Celso B, Tepas J, Langland-Orban B, et al. A systematic review and meta-analysis comparing outcome of severely injured patients treated in trauma centers following the establishment of trauma systems. J Trauma 2006;60:371–378; discussion 378.
6. Twijnstra MJ, Moons KG, Simmermacher RK, et al. Regional trauma system reduces mortality and changes admission rates: A before and after study. Ann Surg 2010;251:339–343.
7. Sampalis JS, Denis R, Fréchette P, et al. Direct transport to tertiary trauma centers versus transfer from lower level facilities: Impact on mortality and morbidity among patients with major trauma. J Trauma 1997;43:288–95; discussion 295.
8. Newgard CD, Zive D, Holmes JF, et al.; WESTRN investigators. A multisite assessment of the American College of Surgeons Committee on Trauma field triage decision scheme for identifying seriously injured children and adults. J Am Coll Surg 2011;213:709–721.
9. Sampalis JS, Denis R, Lavoie A, et al. Trauma care regionalization: A process-outcome evaluation. J Trauma 1999;46:565–79; discussion 579.
10. Rubenson Wahlin R, Ponzer S, Lövbrand H, et al. Do male and female trauma patients receive the same prehospital care?: An observational follow-up study. BMC Emerg Med 2016;16:6.
11. Gomez D, Haas B, de Mestral C, et al. Gender-associated differences in access to trauma center care: A population-based analysis. Surgery 2012;152:179–185.
12. NHTSA. Appendix 3: Patient acuity definitions. National EMS Core Content. Available at: https://www.ems.gov/pdf/education/EMS-Education-for-the-Future-A-Systems-Approach/National_EMS_Core_Content.pdf. Accessed October 11, 2021.
13. Shafi S, de la Plata CM, Diaz-Arrastia, et al. Ethnic disparities exist in trauma care. J Trauma Acute Care Surg 2007;63:1138–1142.
14. Tung EL, Hampton DA, Kolak M, et al. Race/ethnicity and geographic access to urban trauma care. JAMA Netw Open 2019;2:e190138.
15. Adapted from National Academies of Science, Engineering, and Medicine (NASEM). A national trauma care system: integrating military and civilian trauma systems to achieve zero preventable deaths. 2016. Available at: https://www.nationalacademies.org/hmd/Reports/2016/A-National-Trauma-Care-System-Integrating-Military-and-Civilian-Trauma-Systems.aspx. Accessed October 11, 2021.
16. York Cornwell E, Currit A. Racial and social disparities in bystander support during medical emergencies on US streets. Am J Public Health 2016;106:1049–1051.
17. Young MF, Hern HG, Alter HJ, et al. Racial differences in receiving morphine among prehospital patients with blunt trauma. J Emerg Med 2013;45:46–52.
18. Haider AH, Weygandt PL, Bentley JM, et al. Disparities in trauma care and outcomes in the United States: A systematic review and meta-analysis. J Trauma Acute Care Surg 2013;74:1195–1205.

19. Rebollo Salazar D, Velez-Rosborough A, DiMaggio C, et al. Race and insurance status are associated with different management strategies after thoracic trauma. J Surg Res 2021;261:18–25.
20. Berry C, Ley EJ, Mirocha J, et al. Race affects mortality after moderate to severe traumatic brain injury. J Surg Res 2010;163:303–308.
21. Gorman E, Frangos S, DiMaggio C, et al. Is trauma center designation associated with disparities in discharge to rehabilitation centers among elderly patients with traumatic brain injury? Am J Surg 2020;219:587–591.
22. Englum BR, Villegas C, Bolorunduro O, et al. Racial, ethnic, and insurance status disparities in use of posthospitalization care after trauma. J Am Coll Surg 2011;213:699–708.
23. Roberts AL, Gilman SE, Breslau J, et al. Race/ethnic differences in exposure to traumatic events, development of post-traumatic stress disorder, and treatment-seeking for post-traumatic stress disorder in the United States. Psychol Med 2011;41:71–83.
24. Stolbach BC, Anam S. Racial and ethnic health disparities and trauma-informed care for children exposed to community violence. Pediatr Ann 2017;46:e377–e381.
25. Bode AD, Singh M, Andrews JR, et al. Racial and gender disparities in violent trauma: Results from the NEMSIS database. Am J Emerg Med 2019;37:53–55.
26. Ornato JP, Craren EJ, Nelson NM, et al. Impact of improved emergency medical services and emergency trauma care on the reduction in mortality from trauma. J Trauma 1985;25:575–579.
27. Newgard CD, Fu Rongwei, Lerner EB, et al. Role of guideline adherence in improving field triage. Prehosp Emerg Care 2017;21:545–555.
28. Newgard CD, Nelson MJ, Kampp M, et al. Out-of-hospital decision making and factors influencing the regional distribution of injured patients in a trauma system. J Trauma 2011;70:1345–1353.
29. Newgard CD, Fu R, Bulger E, et al. Evaluation of rural versus urban trauma patients served by 9-1-1 emergency services. JAMA Surg 2017;152:11–18.
30. Hsia RY, Wang E, Saynina O, et al. Factors associated with trauma center use for elderly patients with trauma: a statewide analysis, 1999-2008. Arch Surg 2011;146:585–592.
31. Chang DC, Bass RR, Cornwell EE, et al. Undertriage of elderly trauma patients to state-designated trauma centers. Arch Surg 2008;143:776–781; discussion 782.